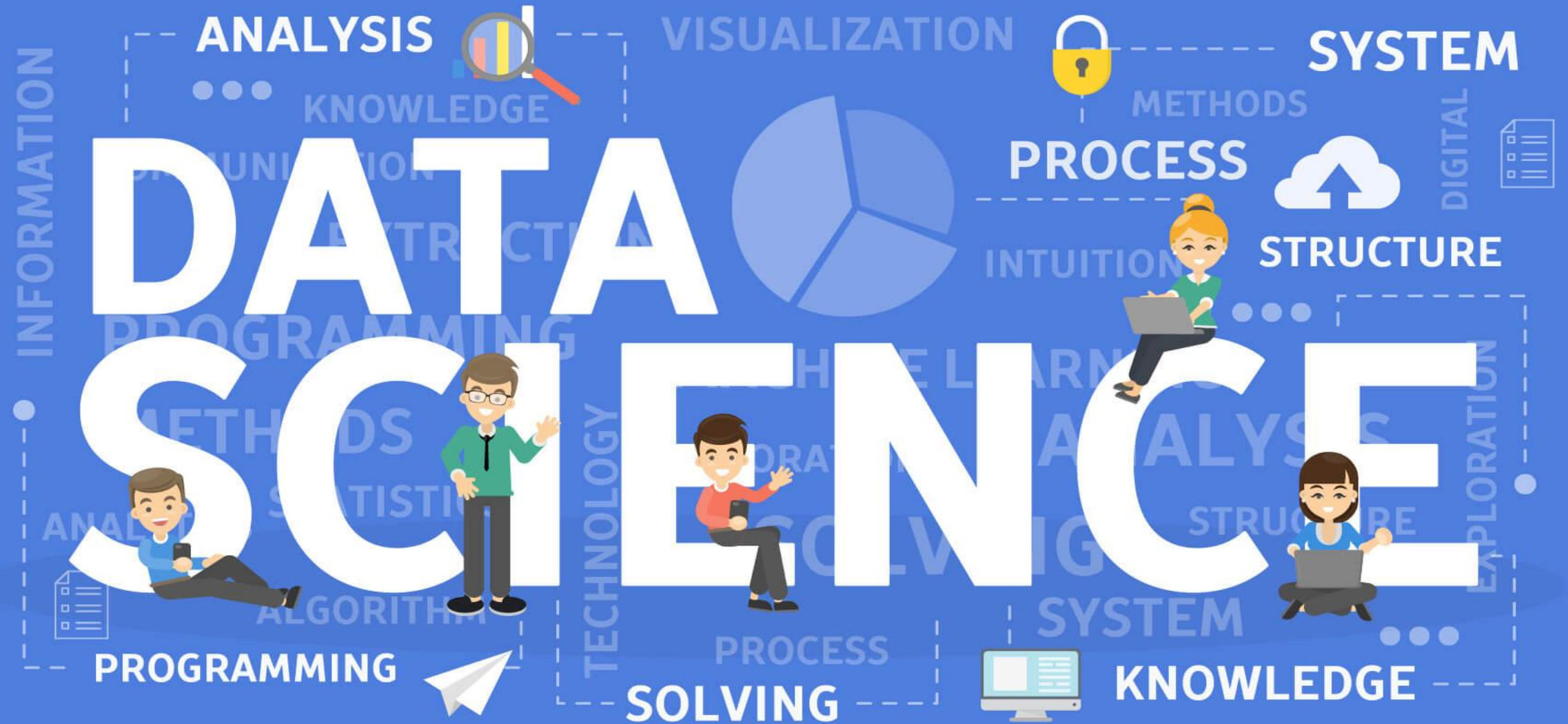
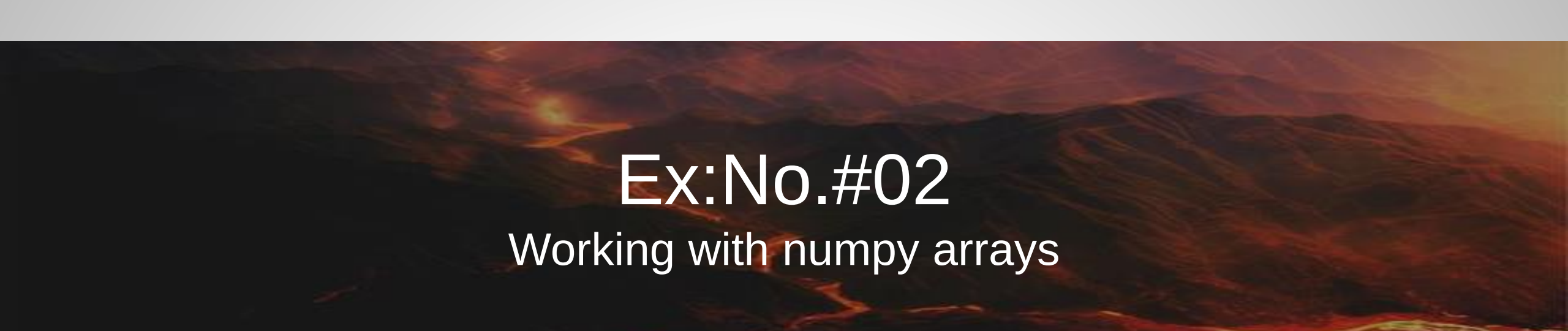






Ex:No.#02

Working with numpy arrays

Procedure : Working with NumPy Arrays

Steps:

1. Open Jupyter Notebook, JupyterLab, or any Python IDE (such as VS Code or PyCharm).

2. Import the NumPy library using:

```
import numpy as np
```

3. Create one-dimensional (1D) and two-dimensional (2D) NumPy arrays using the `np.array()` function.

4. Display the arrays and examine their properties such as shape, size, dimensions (`ndim`), and data type (`dtype`).

5. Access individual elements to extract subsets of data using `slicing`.

6. Perform arithmetic operations such as addition, subtraction, multiplication, and division on arrays.

8. Apply statistical and aggregate functions such as `sum()`, `mean()`, `max()`, `min()`, and `std()`.

10. Join multiple arrays using `concatenate()` and split arrays using `array_split()`.

11. Sort array elements using `np.sort()` and filter elements based on conditions using Boolean indexing.

12. Execute the program and verify the output for each operation.

13. Analyze the results how NumPy arrays efficiently store, manipulate, and process numerical data.

14. Save the notebook or Python file for future reference.

Working with numpy(Numerical Python) Arrays

Creating numpy Arrays

```
import numpy as np

# 1D Array
a = np.array([10, 20, 30, 40, 50])
print(a)

# 2D Array
b = np.array([[1, 2, 3],
              [4, 5, 6]])
print(b)
```

Check Array Properties

```
import numpy as np

arr = np.array([[1, 2, 3], [4, 5, 6]])

print("Shape:", arr.shape)
print("Dimensions:", arr.ndim)
print("Size:", arr.size)
print("Data Type:", arr.dtype)
```

Working with numpy(Numerical Python) Arrays - Accessing

Single Dimensional

```
import numpy as np

# 1D Array
a = np.array([10, 20, 30, 40, 50])
print(a)

# 2D Array
b = np.array([[1, 2, 3],
              [4, 5, 6]])
print(b)
```

Two Dimensional

```
import numpy as np

arr = np.array([[10, 20, 30],
                [40, 50, 60]])

print(arr[0, 1])
print(arr[1, 2])
```

Modifying Arrays

```
import numpy as np

arr = np.array([10, 20, 30])

arr[1] = 100

print(arr)
```

Working with numpy(Numerical Python) Arrays - Operations

Mathematical Operations

```
import numpy as np

a = np.array([1,2,3])
b = np.array([4,5,6])

print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Aggregate Functions

```
import numpy as np

arr = np.array([10,20,30,40,50])

print("Sum:", np.sum(arr))
print("Mean:", np.mean(arr))
print("Maximum:", np.max(arr))
print("Minimum:", np.min(arr))
print("Standard Deviation:", np.std(arr))
```

Working with numpy(Numerical Python) Arrays - Operations

Join Arrays

```
import numpy as np

a = np.array([1,2,3])
b = np.array([4,5,6])

c = np.concatenate((a,b))

print(c)
```

Filtering

```
import numpy as np

arr = np.array([10,20,30,40,50])

result = arr[arr > 25]
print(result)
```

Split Arrays

```
import numpy as np

arr = np.array([10,20,30,40,50,60])

result = np.array_split(arr, 3)

print(result)
```

Slicing Arrays

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50, 60])

print(arr[1:4])
print(arr[:3])
print(arr[3:])
```

Working with numpy(Numerical Python) Arrays -

Students Mark Analysis

```
import numpy as np

marks = np.array([85, 90, 78, 88, 92])

print("Marks:", marks)
print("Average:", np.mean(marks))
print("Highest:", np.max(marks))
print("Lowest:", np.min(marks))

bonus = marks + 5
print("Bonus Marks:", bonus)

passed = marks[marks >= 80]
print("Students Scored Above 80:",
      passed)
```

Sorting

```
import numpy as np

arr = np.array([50,10,40,20,30])

print(np.sort(arr))
```

Feature	Description
Linear Algebra	Create a 2D NumPy array of size 3×3 containing the numbers 1 to 9
	Create a 3×2 array filled with zeros.
	Create a 2×4 array filled with ones.
	Addition , Multiplication of Matrix
	Create a 5×5 matrix with values from 1 to 25.
	Extract the diagonal elements of the matrix.
	Replace all even numbers with 0.
	Find the sum of each row and each column.
Solving Linear Equations	Solve the following system of equations using NumPy: $2x + 3y = 8$ $4x + y = 10$ Display the values of x and y.